# Topspin as an Aerodynamic Stabiliser: A Physics Simulation and Probabilistic Analysis of Table Tennis Ball Flight

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## Abstract

This paper asks how accurately the flight of a table tennis ball can be predicted from classical physics once probability is taken into account, and how much of that predictability comes from spin. A deterministic model built on a saturating Magnus lift law, a Reynolds-dependent drag coefficient, and exponential spin decay produces the baseline trajectories. A Monte Carlo layer then adds Gaussian air turbulence and player execution error, turning each shot into a distribution of possible landing points. Topspin emerges from this analysis as a passive aerodynamic stabiliser. It steepens the descent angle by $3.2^\circ$, raises the maximum safe launch speed by 61% at 6000 RPM, cuts the mean squared landing displacement by 81%, and holds a 91% shot-success rate under moderate execution error where a flat drive manages only 45%. These effects are not separate: all of them follow from a single property, namely that topspin contracts the sensitivity of landing position to launch error by 62%. A three-dimensional extension reproduces curved sidespin serves, and an optimisation locates the error-minimising spin at each simulated launch speed. To close the loop, the model is tested against 133 real recorded trajectories from a public robot dataset, and it predicts the analysed shots to a median per-trajectory error of under four centimetres from their measured launch and spin. The results put numbers behind the familiar coaching instruction to prioritise topspin, and they leave behind a stochastic trajectory framework that carries over to other rotating-ball sports once each ball's own drag and lift coefficients are supplied.

**Keywords:** Table Tennis; Magnus Force; Monte Carlo Simulation; Aerodynamics; Stochastic Modelling; Sports Physics.

## Acknowledgement

This project was designed and carried out by Shloka Maruthi. An earlier baseline version of this simulation engine was presented at SIMULTECH 2026, the 16th International Conference on Simulation and Modeling Methodologies, Technologies and Applications (Porto, Portugal, July 2026); the present paper extends that work by introducing stochastic Euler–Maruyama turbulence, a three-dimensional sidespin extension, execution-error tolerance modelling, and empirical validation against recorded dataset trajectories. I thank an experienced table tennis coach for practical insight that informed the coaching interpretation in Section 5, and Prof. Ganesh Mani for research mentorship. The real-trajectory test in Section 4.15 uses the publicly available SpinDOE dataset (Gossard et al., 2023). I used Claude (Anthropic), a large language model, for code debugging, language polishing and grammar error checking. All scientific content, simulation code, analysis, figures, and conclusions are my own.

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# 1 Introduction

Table tennis looks simple from across the room: two players knocking a small ball back and forth over a 2.74 m table. The physics of that ball in flight is anything but simple. The ball weighs 2.7 g and measures 40 mm across, light enough that a gentle draught can push it noticeably off line. Strong players lean on heavy topspin, delivered through a topspin loop or loop drive, to keep fast shots dropping onto the table. Beginners tend to hit flat, and their shots either sail long or dive into the net. This paper works out why, and what the physics says about playing better.

A useful way to picture a spinning table tennis ball is as a small aircraft on final approach to a runway. The comparison is more than decorative, and the rest of this section makes it concrete. An aircraft trims its descent with elevator surfaces; topspin does the same job through the Magnus force, which bends the trajectory down toward the table. An aircraft holds its heading against gusts with a vertical stabiliser and the rigidity of its spinning gyroscopes; the angular momentum of a spinning ball keeps its spin axis, and therefore the direction of the Magnus force, pointing in much the same direction through the flight, the same gyroscopic stiffness that resists tilting a spinning top. Fly-by-wire aircraft use onboard computers to smooth out small pilot errors; topspin is a passive, purely mechanical analogue, quietly reducing how much small mistakes in launch angle or speed move the landing point. Section 4 attaches numbers to all three of these.

The central question is straightforward: how accurately can I predict where a table tennis ball lands, and how much does spin help? To answer it, a physics simulation was written in Python from the underlying aerodynamic equations, then extended into a probabilistic model with Monte Carlo methods, running each shot hundreds of times with small random disturbances to map the spread of landing points.

## 1.1 Research Objectives

The study has three aims. The first is to build a physically motivated aerodynamic model from a spin-dependent Magnus lift law, a velocity-dependent drag coefficient, and exponential spin decay. The second is to extend that model into a stochastic one with Euler–Maruyama integration, so that turbulence and human execution error enter probabilistically rather than as fixed offsets. The third is to quantify the stabilising effect of topspin across four complementary measures: the safe-speed corridor, descent angle, landing spread, and tolerance to human error.

## 1.2 Prior Work

Wind tunnel work on spinning spheres established that the Magnus lift coefficient depends on the ratio of surface spin speed to flight speed (Mehta, 1985), and the Magnus force on smooth rotating spheres has been measured directly (Barkla and Auchterlonie, 1971). Watts and Ferrer confirmed this spin-parameter dependence directly for baseballs (Watts and Ferrer, 1987), and the same picture has been applied to spinning balls in other sports (Mehta and Pallis, 2001). For table tennis in particular, Miyazaki et al. (2017) recovered the drag and lift coefficients of a spinning ball from flight experiments across the speed and spin ranges relevant to table-tennis shots. Their measured coefficients inform the magnitudes used here, while their finding of a non-monotonic "lift crisis" at intermediate spin marks a limit of any smoothly saturating lift law, a point I return to in Section 6.1. Exponential spin decay from aerodynamic friction during flight has been characterised for sports balls (Cross, 2011); leaving it out makes a simulation overstate the Magnus force late in the trajectory. The stochastic layer follows standard treatments of the Euler–Maruyama method for stochastic differential equations (Lord et al., 2014) and of the Fokker–Planck description of spreading probability clouds (Risken, 1989).

# 2 Theoretical Framework

## 2.1 The Equations of Motion

The ball moves in a vertical plane. At every instant three forces act at once: gravity pulling it down, aerodynamic drag opposing its motion, and the Magnus force acting at right angles to its velocity. Newton's second law ties them together:

$$m\ddot{\mathbf{r}} = \mathbf{F}_{\text{grav}} + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{magnus}} \tag{1}$$

The left side is mass times acceleration; the right side is the vector sum of the forces. The simulation integrates this forward in time, computing the acceleration at each instant, updating the velocity, and updating the position, until the ball reaches the table plane. Table 1 lists the physical constants. The mass (2.7 g) and diameter (40 mm) are fixed by ITTF regulations for the modern plastic ball (ITTF, 2023). Air density and viscosity are those of dry air at 15°C at sea level (ISA standard). The spin decay constant is adopted as a single representative value, consistent with the aerodynamic spin-down of sports balls (Cross, 2011).

**Table 1:** Physical constants used in all simulations.

| Symbol | Quantity | Value | Units |
|---|---|---|---|
| $m$ | Ball mass | 0.0027 | kg |
| $r$ | Ball radius | 0.020 | m |
| $d$ | Ball diameter | 0.040 | m |
| $A$ | Cross-section area | $1.257 \times 10^{-3}$ | $\text{m}^2$ |
| $\rho$ | Air density | 1.225 | $\text{kg m}^{-3}$ |
| $\mu$ | Dynamic viscosity | $1.81 \times 10^{-5}$ | Pa s |
| $g$ | Gravity | 9.81 | $\text{m s}^{-2}$ |
| $k_s$ | Spin decay constant | 0.08 | $\text{s}^{-1}$ |

## 2.2 Drag and the Reynolds Number

Drag acts opposite to the ball's velocity and grows with the square of speed:

$$\mathbf{F}_{\text{drag}} = -\tfrac{1}{2}\rho A C_D(\text{Re})\, v^2\, \hat{\mathbf{v}} \tag{2}$$

Most earlier models fixed the drag coefficient, usually at $C_D = 0.47$ or 0.55. In reality it varies with speed through the Reynolds number, a dimensionless comparison of inertial to viscous forces:

$$\text{Re} = \frac{\rho v d}{\mu} \tag{3}$$

At typical rally speeds of 8 to 14 $\text{m s}^{-1}$ this gives $\text{Re} \approx 22{,}000$ to 38,000, a regime where the drag coefficient still varies with flow speed. The Reynolds dependence of sphere drag is well established (Achenbach, 1972), although the table-tennis range sits below the very high Reynolds numbers of that classic study; table-tennis-specific flight measurements are given by Miyazaki et al. (2017). I model the variation with a sigmoid:

$$C_D(\text{Re}) = 0.55 - \frac{0.08}{1 + e^{\,1.5\times10^{-4}(\text{Re}-28000)}} \tag{4}$$

This runs from $C_D \approx 0.47$ at low speed to 0.55 at high speed. This is a deliberately simplified parameterisation: flight measurements of real table tennis balls give drag coefficients that vary with Reynolds number and sit somewhat lower in parts of this range (Miyazaki et al., 2017), so the curve is an approximate representation rather than a direct fit. A fixed coefficient overstates drag at the lower end, and that error accumulates over the full flight.

## 2.3 The Magnus Force

A spinning ball drags air around with it. On one side the airflow speeds up and on the other it slows, and the resulting pressure difference produces a force at right angles to the velocity. This is the Magnus force. For a topspin loop it points downward and curves the flight toward the table. The same effect swings a cricket ball, bends a sliced golf shot sideways, and produces the banana kick in football.

$$\mathbf{F}_{\text{magnus}} = \tfrac{1}{2}\rho A\, C_L(S)\, v^2\, \hat{\mathbf{n}} \tag{5}$$

where $\hat{\mathbf{n}} = (\boldsymbol{\omega} \times \mathbf{v})/|\boldsymbol{\omega} \times \mathbf{v}|$ sets the force direction. The lift coefficient $C_L$ depends on the spin parameter $S$, the ratio of the ball's surface speed from spinning ($\omega r$) to its flight speed ($v$):

$$S = \frac{\omega r}{v}, \qquad C_L(S) = \frac{1}{2 + 1/S} \tag{6}$$

Wind tunnel work established that this lift coefficient depends mainly on $S$ (Watts and Ferrer, 1987). For its functional form I adopt a simple saturating parameterization, chosen so that lift vanishes at zero spin and levels off toward $C_L = 0.5$ at high spin rather than growing without bound. The spin parameter used here, $S = \omega r/v$, is the same quantity as the dimensionless spin rate $\pi d n/v$ (with $n$ the spin in revolutions per second) measured by Miyazaki et al. (2017), so their range $0 < S < 1$ maps directly onto the lower and middle of the table-tennis range examined here; the highest-spin cases ($S$ up to about 1.5) extrapolate modestly beyond it. Over $S \approx 0.3$ to 1.0 the parameterization gives $C_L$ between roughly 0.2 and 0.4, comparable in magnitude to their measurements, though it deliberately smooths over the non-monotonic lift crisis they observed near $S = 0.5$ (Section 6.1). A purely linear law, $C_L = 1.2S$, keeps climbing without bound, so extrapolated to the high spin of a loop it overstates the lift: fed into the trajectory of Section 4.3 it predicts a descent-angle change of about $12^\circ$ against the $3.2^\circ$ of the saturating form, close to four times larger. Capping the lift keeps the geometry realistic; the resulting trajectory model as a whole, rather than the lift law in isolation, is what I assess against recorded shots in Section 4.15.

## 2.4 Spin Decay

Air resistance slowly bleeds off the ball's rotation during flight, modelled as exponential decay:

$$\omega(t) = \omega_0\, e^{-k_s t} \tag{7}$$

This exponential form is supported directly for table tennis: Ziegler et al. (2023) measured an in-flight spin-damping coefficient of $k = 0.091 \pm 0.03\,\text{s}^{-1}$ from tracked shots, in line with the aerodynamic spin-down characterised for other sports balls (Cross, 2011). I use $k_s = 0.08\,\text{s}^{-1}$, which lies within that measured range; the ball then keeps about 97.6% of its initial spin over a typical 0.3 s flight ($e^{-0.08\times 0.3} \approx 0.976$). The effect is small on that timescale but is kept for completeness, and it grows for slower or longer trajectories such as a floated backspin push or chop. In the simulation the decay is applied at every timestep, $\omega_{t+dt} = \omega_t\, e^{-k_s\, dt}$.

## 2.5 Stochastic Turbulence Model

Real flights are never perfectly smooth. Random air currents nudge the ball at every instant. I model this with the Euler–Maruyama method (Lord et al., 2014), adding Gaussian noise to the velocity at each timestep:

$$\mathbf{v}_{t+dt} = \mathbf{v}_t + \mathbf{a}_{\text{det}}\, dt + \sigma\sqrt{dt}\;\boldsymbol{\xi}, \qquad \boldsymbol{\xi} \sim \mathcal{N}(0, \mathbf{I}) \tag{8}$$

Here $\boldsymbol{\xi}$ is an independent standard normal vector drawn at each step, and the parameter $\sigma$ sets the turbulence strength. Because the noise is added to the velocity, and $\sigma\sqrt{dt}$ must carry units of velocity, $\sigma$ has units of

$\mathrm{m\,s^{-3/2}}$. The $\sqrt{dt}$ factor makes the noise scale as Brownian motion, with small random steps that build up naturally over time rather than growing or shrinking with the choice of timestep (Risken, 1989). I check this timestep invariance directly (Section 3): the landing statistics are unchanged when $dt$ is halved. The formulation links ball flight to the Fokker–Planck equation, which describes how a probability cloud spreads under random forcing. The effective velocity-diffusion coefficient is proportional to $\sigma^2$ and sets how fast the landing distribution widens under the random forcing; for a fixed forcing strength, topspin narrows that distribution by stabilising the direction of travel.

### 2.6 Dimensional Analysis

Before the results it is worth asking which dimensionless quantities really govern the flight, since those decide how far the conclusions reach beyond a table tennis ball. The landing distance $x_{\text{land}}$ depends on launch speed $v$, spin $\omega$, ball radius $r$, air density $\rho$, viscosity $\mu$, gravity $g$, and mass $m$. These carry three independent dimensions, so by the Buckingham $\pi$ theorem (Buckingham, 1914) the eight quantities collapse into $8-3=5$ dimensionless groups. Taking $v$, $r$, and $\rho$ as repeating variables gives

$$\frac{x_{\text{land}}}{r} = F\left(S = \frac{\omega r}{v},\ \mathrm{Re} = \frac{\rho v d}{\mu},\ \mathrm{Fr} = \frac{v^2}{gd},\ \Pi_m = \frac{m}{\rho d^3}\right) \tag{9}$$

The first two are the spin parameter and Reynolds number already introduced. The third is a Froude number comparing inertia to gravity, and the fourth is the ratio of the ball's mass to the mass of the air it displaces. Every trajectory in this paper is a point in this four-dimensional space. The practical upshot is that the method is not specific to table tennis. Any rotating sphere that matches these four groups traces the same nondimensional path, so the framework and the qualitative stabilisation mechanism carry over to other rotating-ball sports such as tennis, cricket, and football. The exact numbers do not carry over directly, because seams, fuzz, and surface panels give each of those balls its own drag and lift curves; what transfers is the modelling approach and the mechanism, once those ball-specific coefficients are put in. Within this picture, $S$ is the single group a player controls directly through spin.

## 3 Methods

### 3.1 Python Implementation

All simulations were written in Python with NumPy, SciPy, and Matplotlib. The physics lives in one engine class, so every experiment shares the same equations, constants, and coordinate frame (table surface at height zero). A condensed excerpt of the core acceleration routine, which returns the time-derivative of the velocity, is shown below.

```
def accel(x, y, vx, vy, t, omega0, s):
    v  = hypot(vx, vy)
    Re = RHO * v * D / MU
    Cd = 0.55 - 0.08/(1 + exp(1.5e-4*(Re-28000)))
    omega = omega0 * exp(-KS * t)   # spin decay
    S  = omega * R / v              # spin parameter
    Cl = 0.0 if S < 1e-12 else 1.0/(2.0 + 1.0/S)  # saturating, ->0.5
    kD = 0.5*RHO*A*Cd*v / M         # drag / mass
    kM = 0.5*RHO*A*Cl*v / M         # Magnus / mass
    ax = -kD*vx + s*kM*vy   # s=+1 topspin, -1 backspin
    ay = -kD*vy - s*kM*vx - G
    return ax, ay
```

Deterministic trajectories are integrated with a classical fourth-order Runge–Kutta scheme at $dt = 0.001$ s. Turbulent trajectories use Euler–Maruyama (Section 3.3), since a Wiener-noise term is only first-order integrable. Before any figure is produced, the engine passes a set of physical self-tests that establish its correctness. In the zero-air, zero-spin limit it reproduces the analytic projectile range $v^2 \sin 2\theta / g$ to machine precision. Topspin lands short and steep while backspin lands long and shallow. $C_L$ stays below its 0.5 ceiling and rises monotonically with $S$, and $C_D$ stays inside $[0.47, 0.55]$ across match speeds. The retained spin matches $e^{-k_s t}$, mechanical energy falls monotonically under drag as it must, and the stochastic landing spread is unchanged when the timestep is halved. The three-dimensional engine reduces to the two-dimensional one to machine precision at zero sidespin. All Monte Carlo experiments use a fixed random seed, so every reported figure is reproducible under the same implementation.

### 3.2 Deterministic Baseline

For clean baseline trajectories the stochastic noise term in Equation 8 is set to zero and the system is integrated with Runge–Kutta. This gives the ideal trajectory for a set of launch parameters, and serves as the reference against which the probabilistic results are compared.

### 3.3 Monte Carlo Stochastic Runs

For the turbulence experiments the Euler–Maruyama method is used rather than Runge–Kutta. It is a standard, computationally simple scheme for the additive-noise stochastic differential equation used here; because the noise amplitude $\sigma$ is constant, the higher-order Milstein correction vanishes, so a first-order scheme is appropriate (Lord et al., 2014), and the timestep-halving test of Section 3.1 confirms it is sufficient for the reported statistics. Each condition was run 300 to 600 times at three turbulence levels, expressed as velocity-diffusion intensities $\sigma \in \{0.016, 0.063, 0.158\}\,\mathrm{m\,s^{-3/2}}$ (calm, club-hall, and a stiff draught). From each batch I computed the mean trajectory, 95% confidence intervals, the landing histogram, and the mean squared displacement.

### 3.4 Phase Space Sweep

A $40 \times 40$ grid (1 600 shots) covered spin rates 0 to 6 000 RPM and launch speeds 3 to 20 $\mathrm{m\,s^{-1}}$, with the ball struck from near the server's baseline ($x_0 = 0.15$ m) at a contact height of 0.15 m. Each shot was classified as

$$\text{outcome} = \begin{cases} \text{net} & \text{ball hits the net} \\ \text{safe} & x_{\text{net}} < x_{\text{land}} \le x_{\text{table}} \\ \text{long} & x_{\text{land}} > x_{\text{table}} \end{cases} \tag{10}$$

### 3.5 Human Error Tolerance

To model execution mistakes, independent Gaussian noise was added to each launch parameter:

$$v = v_0 + \mathcal{N}(0, \lambda^2 \cdot 1.5^2) \tag{11}$$

$$\theta = \theta_0 + \mathcal{N}(0, \lambda^2 \cdot 1.5^2) \quad \text{(degrees)} \tag{12}$$

$$\omega = \max\bigl(0,\ \omega_0 + \mathcal{N}(0, \lambda^2 \cdot 400^2)\bigr) \tag{13}$$

For each $\lambda \in [0, 1.5]$, 500 trials were run. Two shots were compared: a flat drive at 9 $\mathrm{m\,s^{-1}}$, 5°, 0 RPM, and a topspin loop at 12 $\mathrm{m\,s^{-1}}$, 8°, 3 500 RPM. The two are launched at different speeds on purpose. A flat drive has to be slower than a loop to stay on the table at all, so each is judged at a realistic operating point for its own style.

# 4 Results

## 4.1 Phase Diagram: How Spin Expands the Safe Zone

The phase diagram (Figure 1) maps the outcome of 1 600 shots across a grid of spin rates and launch speeds. Three regions appear: net collisions in red, safe landings in green, and overshoots in blue.

At 0 RPM the safe green band is narrow, spanning only about 2 m s$^{-1}$ of launch speed. With no Magnus force, gravity alone has to bring the ball down: it must be fast enough to clear the net 1.22 m away, yet slow enough to drop before the far edge. That window is genuinely tight. Any upward slip in swing speed sends the ball long, and any downward slip drops it into the net. This is exactly the bind a beginner faces on a fast flat drive, and even strong players feel it under pressure.

As spin increases the Magnus force adds a downward component that the player can tune, and the safe region climbs steadily. The upper boundary rises from 9.2 m s$^{-1}$ at 0 RPM to about 15 m s$^{-1}$ at 6 000 RPM, while the lower boundary climbs more gently as the extra downforce demands a little more speed to still clear the net. The band widens from roughly 2 m s$^{-1}$ to about 5 m s$^{-1}$, more than doubling. That widening is the aerodynamic case for the loop over the drive.

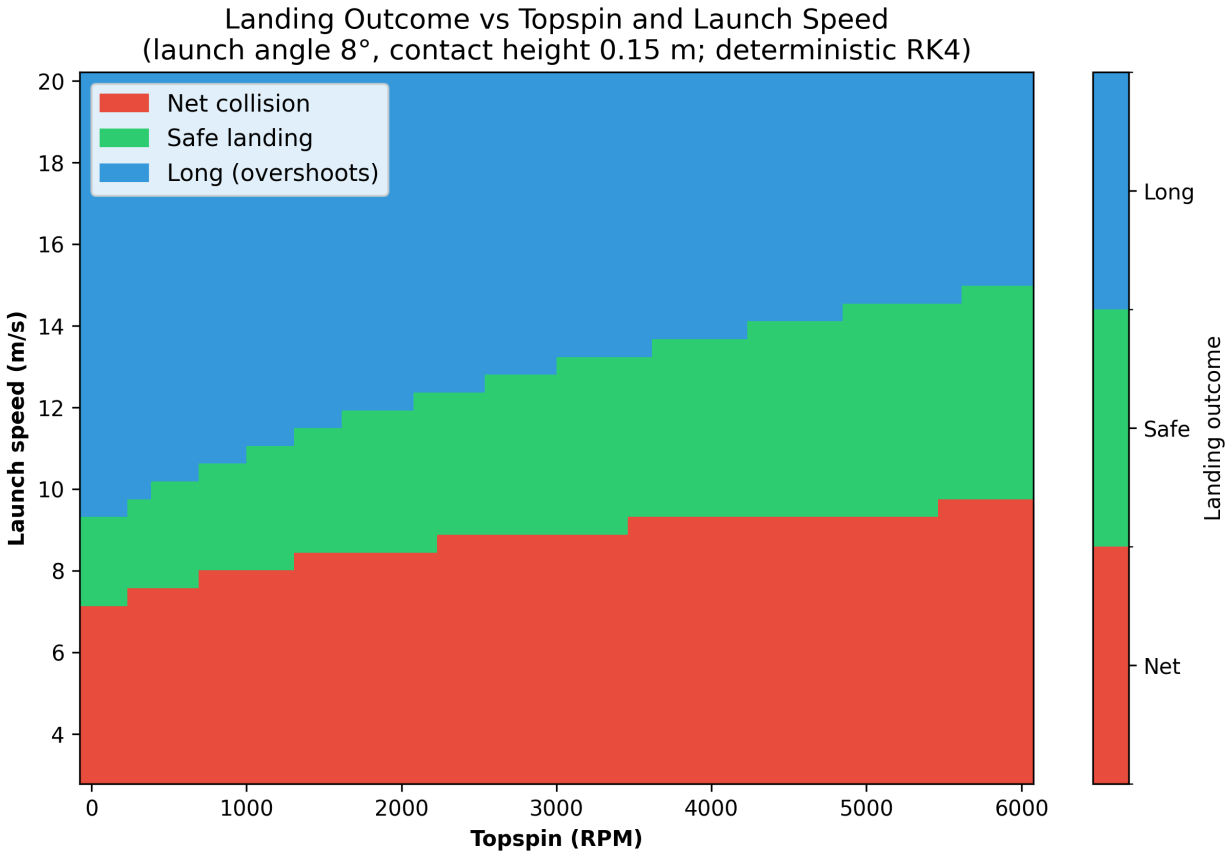



**Figure 1:** Phase diagram of spin rate against launch speed for 1 600 simulated shots (8° launch, contact height 0.15 m). Red: net. Green: safe landing. Blue: overshoot. At 0 RPM the safe band spans only about 2 m s$^{-1}$, the width of a beginner's margin for error. By 6 000 RPM it is more than twice as wide, so a player can vary swing speed substantially and still land the ball. This is the aerodynamic reason coaches teach the loop over the flat drive.

## 4.2 Critical Velocity: The Speed Ceiling

The maximum safe speed at 0 RPM is $9.2\,\mathrm{m\,s^{-1}}$. At 3000 RPM it rises to $13.0\,\mathrm{m\,s^{-1}}$ (+41%), and at 6000 RPM it reaches $14.8\,\mathrm{m\,s^{-1}}$ (+61%), shown in Figure 2. The relationship is smooth and continuous: the safe-speed ceiling rises with spin across the whole range, with no threshold to cross, so the benefit is present from the first revolution. The gain per RPM does shrink at higher spin, in keeping with the diminishing returns seen on the other measures below.

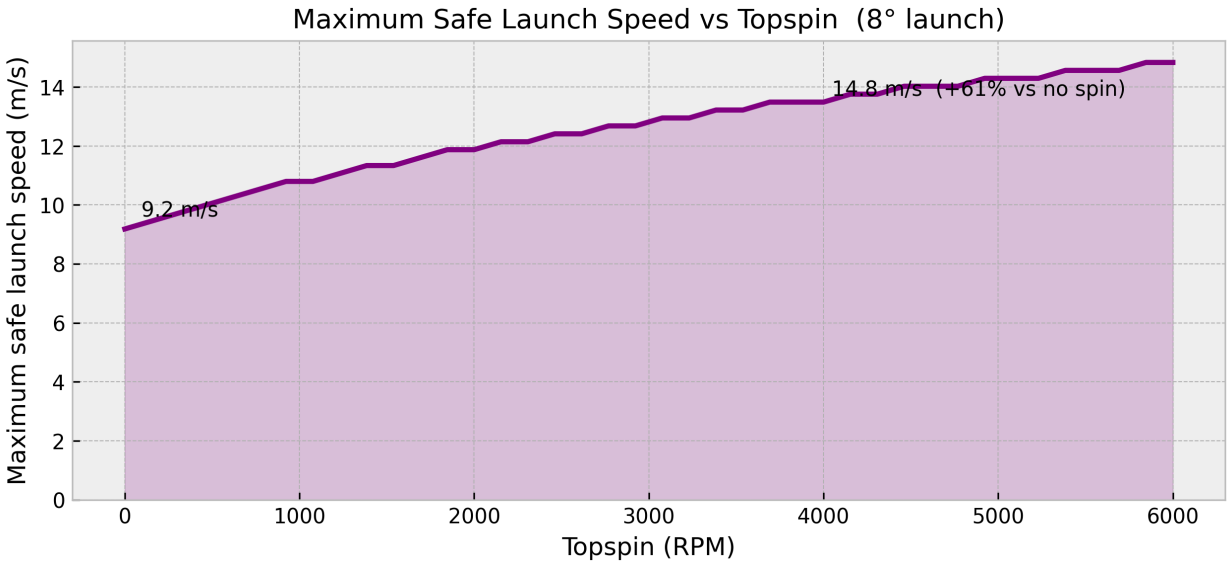



**Figure 2:** Maximum safe launch speed against spin rate (8° launch). At 0 RPM the ceiling is $9.2\,\mathrm{m\,s^{-1}}$. It rises to $13.0\,\mathrm{m\,s^{-1}}$ at 3000 RPM (+41%) and $14.8\,\mathrm{m\,s^{-1}}$ at 6000 RPM (+61%). Every additional RPM from a steeper brush in the loop directly raises how hard the ball can be struck while still landing safely.

## 4.3 Trajectory Shape and Descent Angle

A topspin loop at 3000 RPM steepens the descent angle at landing by 3.2° relative to a flat drive at the same launch (Figure 3). At a shallow descent the ball arrives nearly horizontally, so a few centimetres of extra carry send it off the far end. At a steeper descent the same carry error is far less likely to overshoot. This steepening is the aerodynamic equivalent of an aircraft pitching its nose down onto the instrument-landing-system glideslope, a controlled and repeatable descent onto the target.

A backspin chop or push produces the mirror image. Reversing the sign of $\omega$ reverses the Magnus direction, so instead of pulling the ball down it now partly opposes gravity, giving a descent 3.0° shallower than a flat drive. That explains two familiar features of chop shots: the ball floats low over the net and then drops away near the table. It also explains why attacking a deep backspin push needs extra topspin. The incoming geometry is unfavourable, so the player has to generate more Magnus force to bring the ball back down.

## 4.4 Landing Variance Under Air Turbulence

Figure 4 shows how the spread of landing positions changes with spin under three turbulence levels. Across all three the curves flatten once spin passes about 2000 RPM. The Magnus force has done its main stabilising work by then, and further spin returns little on this particular measure. Stronger turbulence simply lifts the

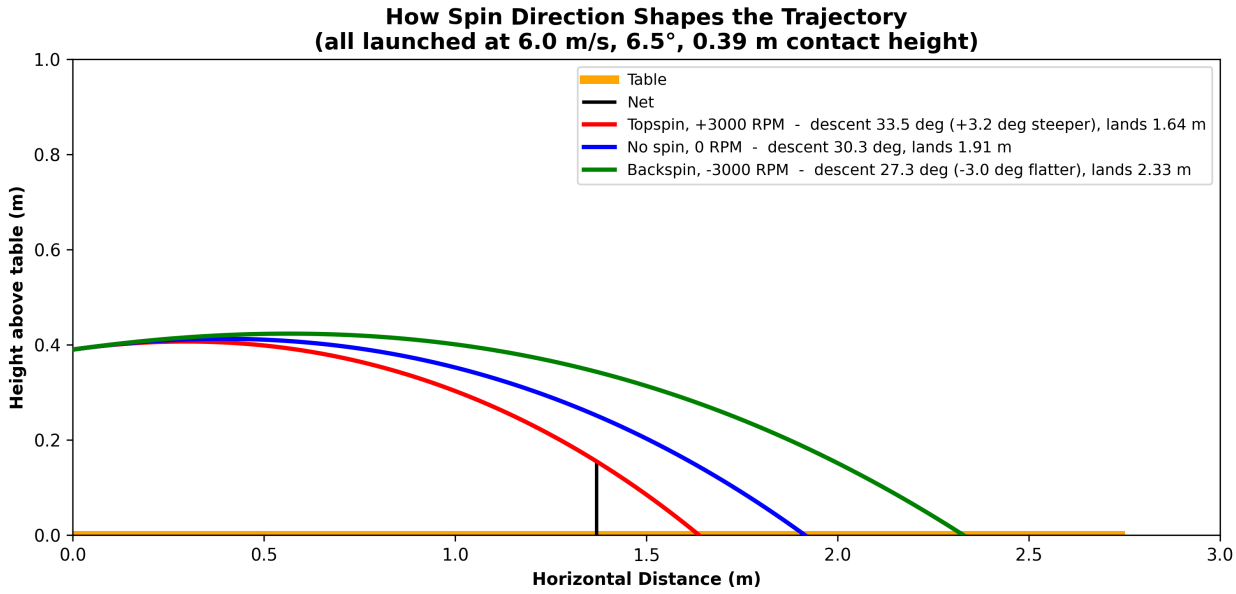



**Figure 3:** Deterministic trajectories for topspin (+3000 RPM), no spin, and backspin (−3000 RPM), all launched at $6.0\,\mathrm{m\,s^{-1}}$, 6.5°. Topspin steepens the descent by 3.2° and lands the ball short and cleanly. Backspin produces the mirror effect, flattening the descent by 3.0° and making the ball float, which is why third-ball attackers must add topspin to overcome the unfavourable incoming geometry.

whole floor. One caveat concerns the left-hand end of each curve: at low spin under strong turbulence most shots overshoot, and only the few knocked short by a large negative fluctuation land in-bounds, so the first plotted points rest on a truncated, self-selected subset that understates the true spread rather than marking genuine low-spin stability. It is worth being precise about what this does and does not show. The stabilising benefit is real and large, but it is captured mostly by a committed loop of a few thousand RPM rather than growing without limit. That is reassuring for a player, because superhuman spin is not required to buy stability, and it fits the coaching intuition that a solid loop, not a maximal one, is what keeps the ball on the table in awkward conditions.

## 4.5 Landing-Position Variance Under Stochastic Forcing

The mean squared displacement of the landing position drops from $7.0 \times 10^{-4}\,\mathrm{m^2}$ at 0 RPM to $1.3 \times 10^{-4}\,\mathrm{m^2}$ at 6000 RPM, an 81% reduction (Figure 5). The stochastic forcing $\sigma$ is identical across all spins, so this 81% drop is not a change in the underlying noise: it measures how much more tightly topspin confines the spread that the noise produces by the time the ball lands. In variance terms the landing-position spread for a heavy loop is more than five times smaller than for a flat drive struck under identical conditions. A player generating heavy topspin is not only aiming in a more controlled direction, but is narrowing the whole probability distribution of where the ball can finish.

## 4.6 Shannon Entropy of Landing Distributions

Standard deviation measures how *wide* a landing distribution is; Shannon entropy measures how *unpredictable* it is. Two distributions can share a width yet differ in shape, one tightly clustered with a few outliers, the other spread out evenly, and entropy captures that difference in a way variance alone cannot. Entropy is computed by sorting centred landing positions into a fixed set of bins and measuring how evenly

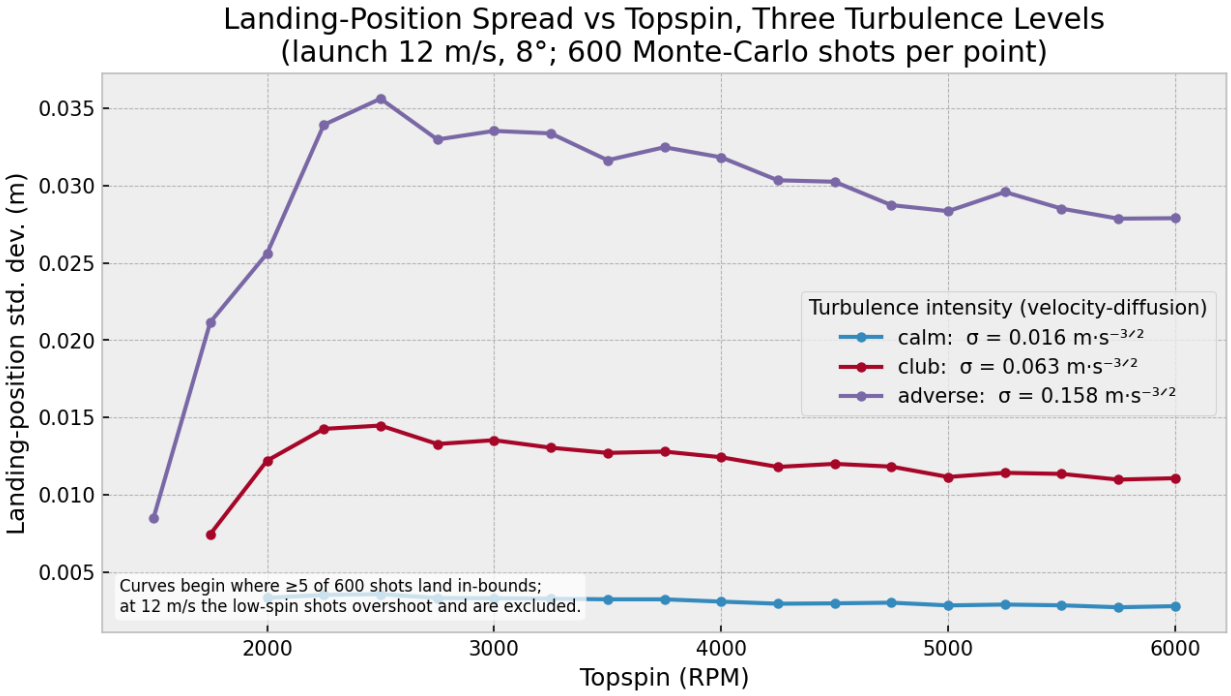



**Figure 4:** Standard deviation of landing position against spin rate at three turbulence levels ($\sigma = 0.016, 0.063, 0.158\,\mathrm{m\,s^{-3/2}}$). The Magnus force does its principal stabilising work by about 2 000 RPM, beyond which the spread flattens at every turbulence level, while stronger turbulence raises the overall floor. Curves begin where at least five of 600 shots land in-bounds; at $12\,\mathrm{m\,s^{-1}}$ the low-spin shots overshoot and are excluded.

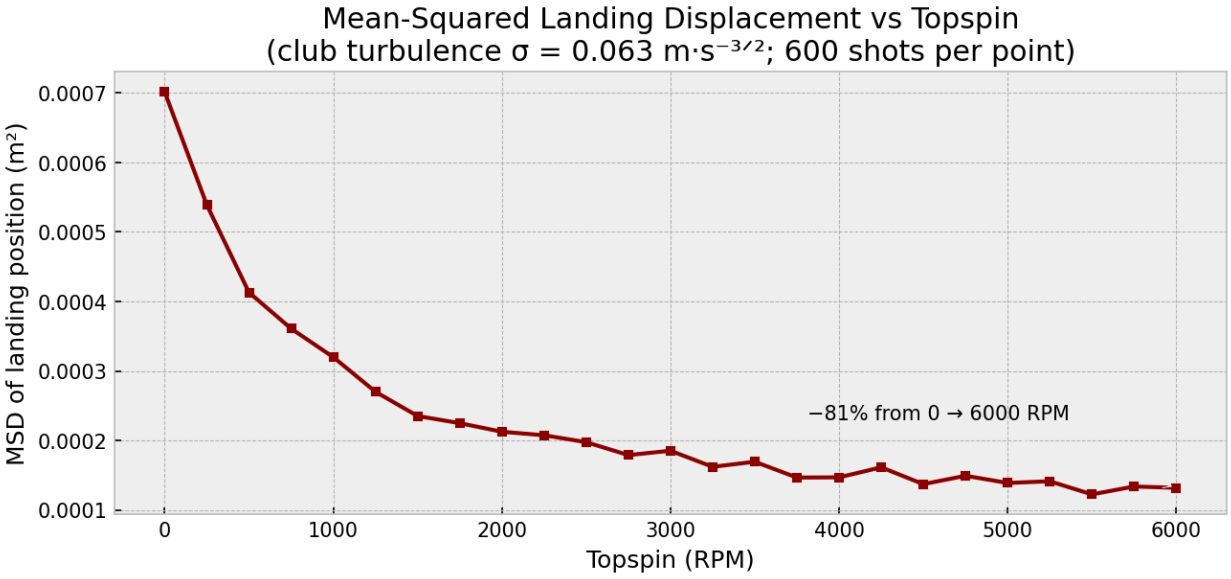



**Figure 5:** Mean squared displacement of landing position against spin rate under club-hall turbulence ($\sigma = 0.063\,\mathrm{m\,s^{-3/2}}$). It drops from $7.0 \times 10^{-4}\,\mathrm{m^2}$ at 0 RPM to $1.3 \times 10^{-4}\,\mathrm{m^2}$ at 6 000 RPM, an 81% reduction. The stochastic forcing is the same at every spin, so this reflects how much more tightly topspin confines the propagated landing spread, more than five times smaller in variance than for a flat drive.

the probability is spread:

$$H(\omega) = -\sum_i p_i(\omega) \log_2 p_i(\omega) \tag{14}$$

The units are bits, and the bin grid is held fixed across all spin rates so the comparison is fair. Table 2 lists the values.

Entropy falls from 3.44 bits at 0 RPM to 2.28 bits at 6000 RPM, a reduction of about 34%, with most of the drop in place by $\approx 4000$ RPM. As a rough reading, $2^H$ falls from about 11 to about 5, so topspin roughly halves the number of effectively distinct landing zones. A flat drive can land in any of about ten plausible zones; a heavy loop concentrates into about five tighter ones. Standard deviation and entropy fall together and monotonically (Table 2), consistent with landing distributions that stay unimodal across the tested spin rates rather than splitting or developing heavy tails.

**Table 2:** Shannon entropy and standard deviation of landing position by spin rate (club-hall turbulence, seeded Monte Carlo).

| Spin (RPM) | $H$ (bits) | $\sigma$ (m) |
|---|---|---|
| 0 | 3.44 | 0.0273 |
| 1000 | 2.81 | 0.0175 |
| 2000 | 2.59 | 0.0147 |
| 3000 | 2.40 | 0.0131 |
| 4000 | 2.39 | 0.0128 |
| 6000 | 2.28 | 0.0117 |

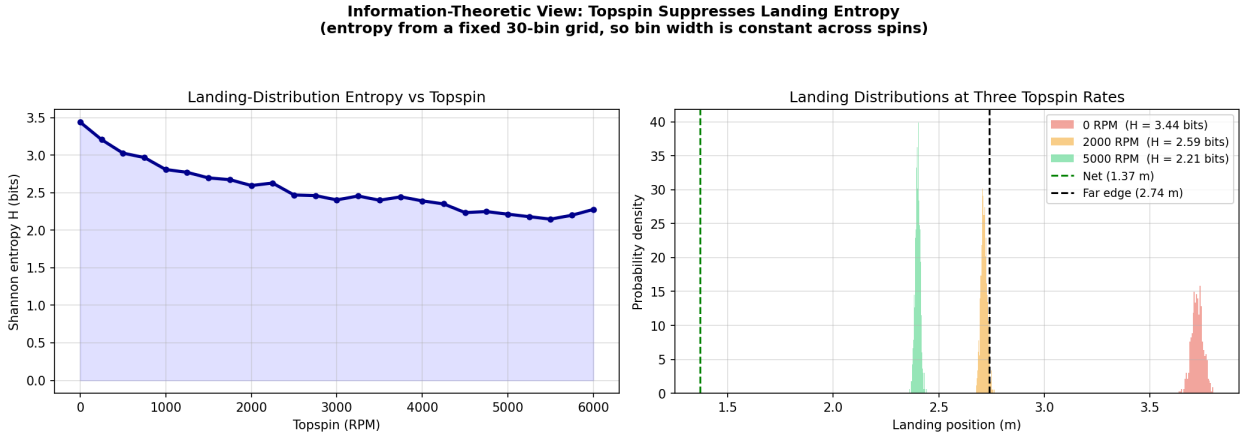



**Figure 6:** Information-theoretic view of landing distributions. Left: entropy against spin rate, falling from 3.44 bits at 0 RPM to 2.28 bits at 6000 RPM, about a 34% reduction (values in Table 2). Right: landing histograms at 0, 2000, and 5000 RPM, showing the distribution concentrating onto the table as spin rises. For an advanced player this translates into placement precision: a loop at 5000 RPM can be aimed at the opponent's elbow with genuine consistency, where a flat drive at the same speed cannot.

## 4.7 Monte Carlo Trajectory Stability

At 12.5 m s$^{-1}$ and 8°, the topspin shot lands on the table 95% of the time with $\sigma = 0.089$ m, while the flat drive lands 0% of the time with $\sigma = 0.167$ m, a variance ratio of 3.5× (Figure 7). The variance ratio actually understates the practical gap, because the flat drive's entire landing distribution falls beyond the far edge of

the table. Under the modelled turbulence, no simulated flat drive at this speed and angle lands on the table: its nominal trajectory already overshoots the far edge, so here the shot needs spin to work at all.

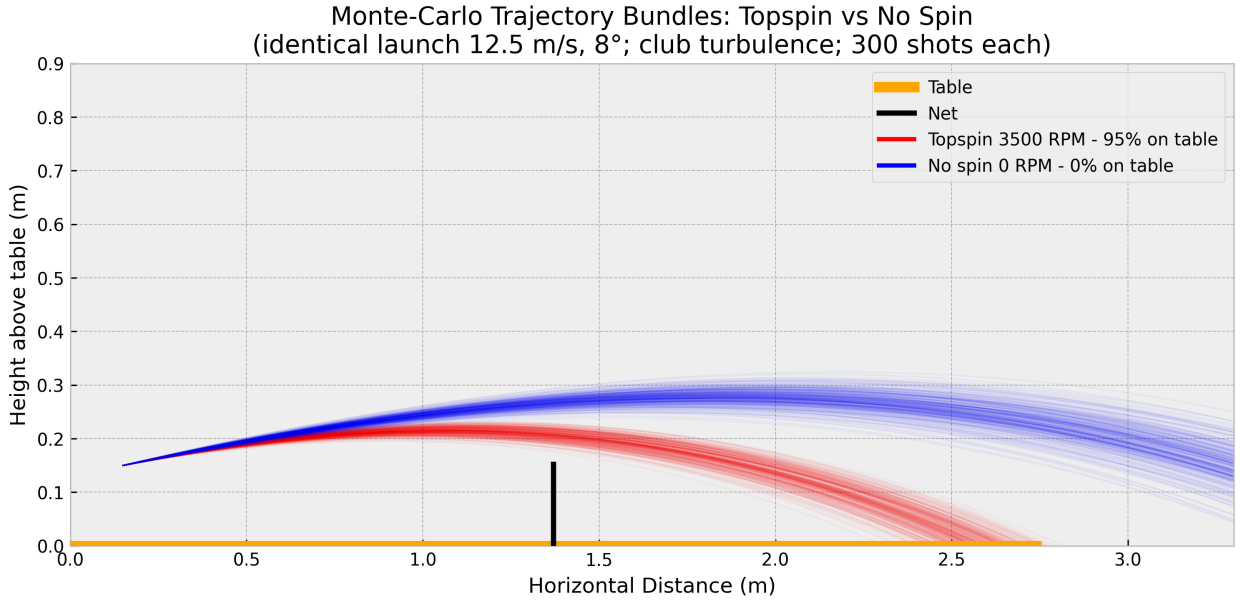



**Figure 7:** Monte Carlo trajectory bundles at $12.5\,\mathrm{m\,s^{-1}}$, 8° (300 shots each). Topspin at 3 500 RPM (red): 95% land on the table, $\sigma = 0.089$ m. Flat drive at 0 RPM (blue): 0% on the table, $\sigma = 0.167$ m, variance ratio 3.5×. The no-spin bundle is not only wider, it sails off the far end in every run. At this speed topspin is the difference between a shot that can work and one that cannot, because the flat drive's nominal trajectory already overshoots the far edge.

## 4.8 Landing Position Distribution

The topspin distribution is tight and sits within the table boundaries, while the flat-drive distribution is shifted entirely past the far end (Figure 8). At this launch speed, spin is not merely helpful; it is what makes the shot work at all.

## 4.9 Launch Geometry: Contact Height and Spin Rate

Figure 9 compares a high contact point ($h = 0.55$ m above the table) with a low one ($h = 0.20$ m) across two shot types, varying spin at a fixed launch angle in each panel. The lower contact point sustains a higher maximum safe speed at every spin rate, and the gap widens as spin increases. The reason is geometric: at a fixed launch angle a ball struck from a lower point has a shorter drop to the table, so it lands sooner and travels less far before reaching the far edge. That lets the player hit harder before the shot overshoots, and because the Magnus force grows with speed, each added RPM does more at those higher speeds.

In the topspin panel the low-contact safe speed rises from about $8.9\,\mathrm{m\,s^{-1}}$ at 0 RPM to about $14.8\,\mathrm{m\,s^{-1}}$ at 6 000 RPM, while the high-contact curve rises only from about 7.0 to $9.2\,\mathrm{m\,s^{-1}}$. Every added RPM is worth more to the low-contact player. The right panel shows the same pattern for near-flat shots at 1°, where the low-contact advantage is even larger, more than twice the safe launch speed at 3 000 RPM.

## 4.10 Human Error Tolerance

Figure 10 shows the probability of a safe landing as a function of execution error, comparing a flat drive with a topspin loop. At error scale $\lambda = 0.5$ the topspin shot succeeds 91% of the time against 45% for the

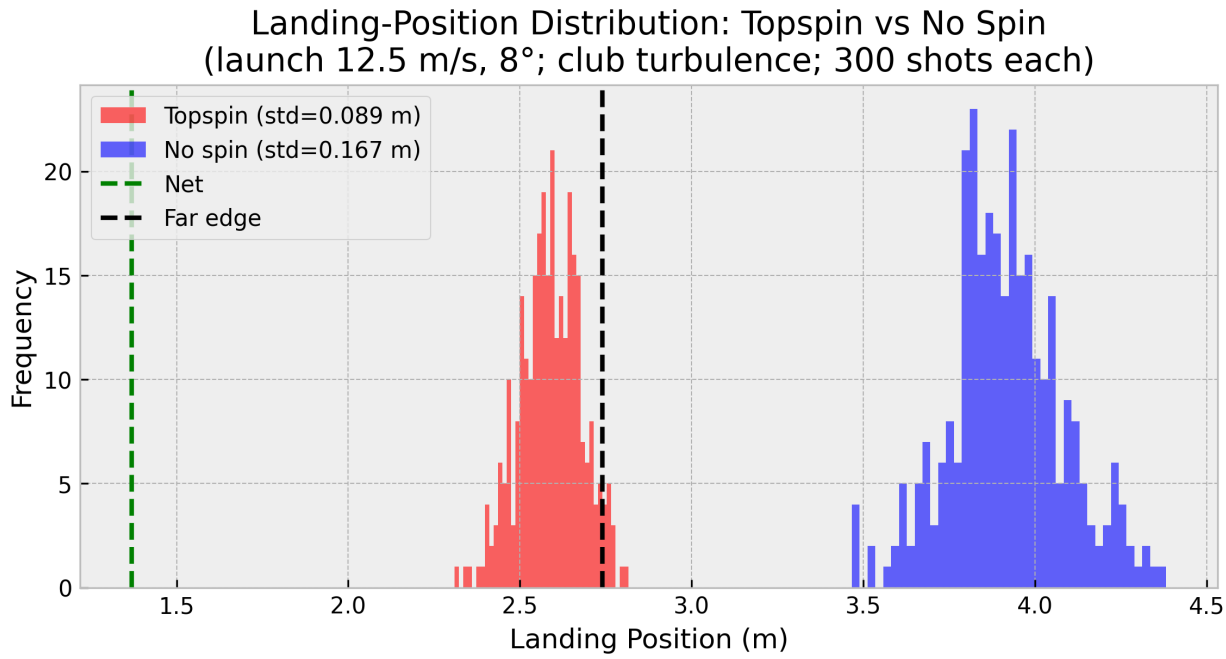



**Figure 8:** Histogram of landing positions across 300 Monte Carlo runs at identical launch speed and angle (12.5 m s$^{-1}$, 8°), differing only in spin. Topspin (red): a tight cluster within the table ($\sigma = 0.089$ m). Flat drive (blue): entirely beyond the far end ($\sigma = 0.167$ m). This shows what a variance ratio of 3.5× means in practice: at this speed the flat drive already overshoots the table, so its landing is out of bounds across the whole distribution.

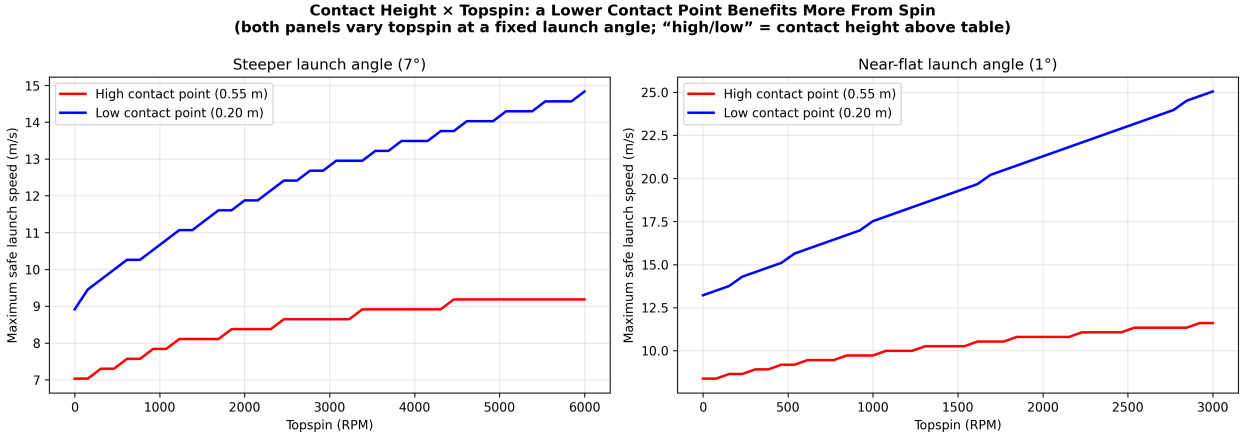



**Figure 9:** Maximum safe launch speed against spin rate for a high contact point ($h = 0.55$ m, red) and a low one ($h = 0.20$ m, blue). Left: steeper 7° launch. Right: near-flat 1° launch. The lower contact point sustains a higher safe speed at every spin rate, and the gap widens with spin. The coaching read is that players taking the ball low and early should prioritise topspin generation, while players contacting from above can lean more on speed and placement.

flat drive, a factor-of-two gap (95% Wilson confidence intervals (Wilson, 1927) from 500 shots: 88 to 93% and 41 to 49%, which do not overlap). At $\lambda = 1.0$ the figures are 66% against 31%, and at $\lambda = 1.5$ they are 48% against 21%.

Every shot carries small execution errors. Even strong players do not repeat a swing perfectly: the contact point shifts by a few millimetres, the racket angle opens or closes a little, the timing runs slightly early or late. A flat drive, played with a square face and no brush, passes all of those errors straight through to the landing point. There is no compensating mechanism. A loop drive generates Magnus force throughout the flight, so when the launch angle is slightly off, the arc adjusts on its own, pulling the ball down a touch sooner or later but still inside the table. The Magnus arc attenuates the error without any input from the player, which is the physical reason coaches teach beginners to loop rather than hit. These two shots differ in speed and angle as well as spin, so this compares two complete shot configurations rather than isolating spin on its own; the clean spin-only comparison is Figure 7, where launch speed and angle are held fixed and only spin differs. Both simulated shots here receive the same error distribution, so the gap between these configurations reflects the flight dynamics rather than execution skill.

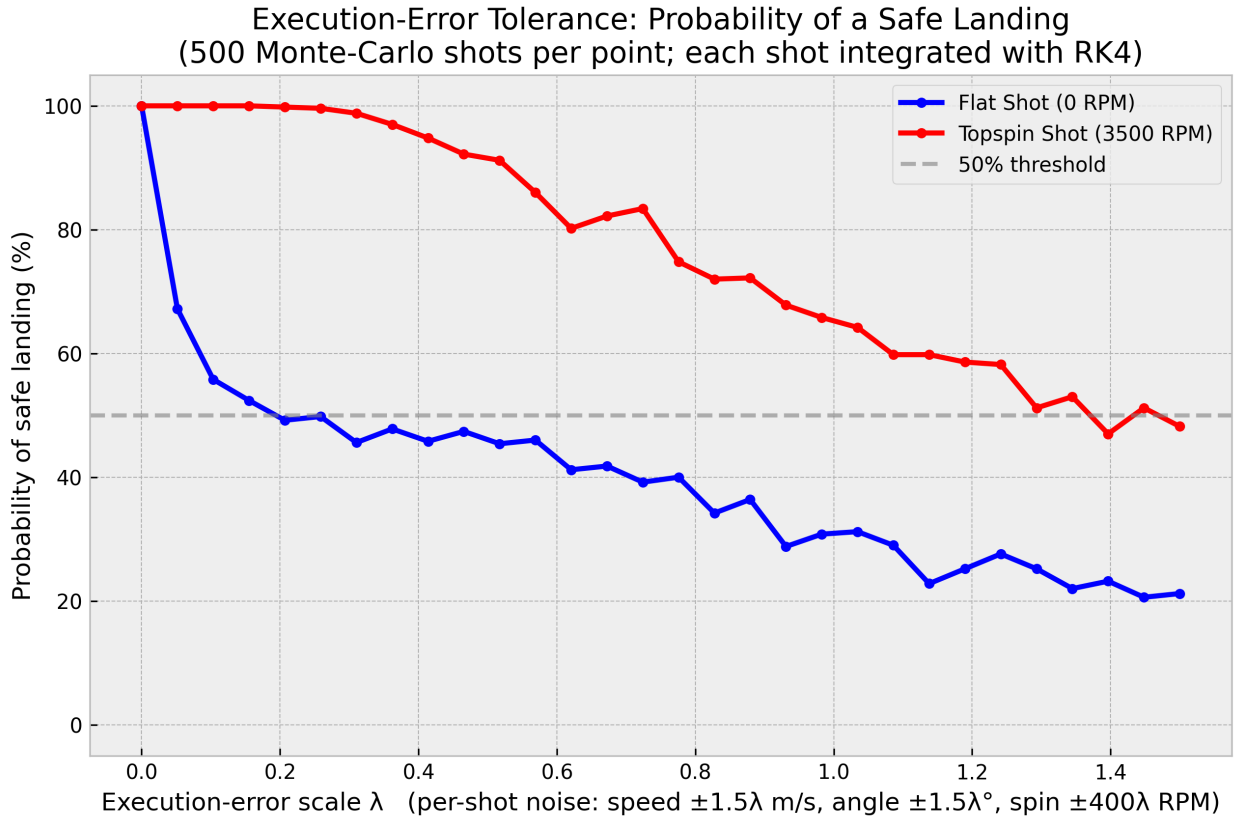



**Figure 10:** Shot success against execution-error scale $\lambda$ for a topspin loop ($12\,\mathrm{m\,s^{-1}}$, 8°, 3 500 RPM) and a flat drive ($9\,\mathrm{m\,s^{-1}}$, 5°, 0 RPM). At $\lambda = 0.5$: 91% against 45%. The topspin shot stays above 50% until $\lambda \approx 1.3$, while the flat drive drops below 50% almost at once. Both shots use identical error distributions; because their speed, angle, and spin all differ, the gap reflects the two complete shot configurations rather than spin alone (Figure 7 isolates spin at fixed speed and angle).

## 4.11 Launch-Error Sensitivity: The Unifying Mechanism

The four results above, a wider safe corridor, a steeper glideslope, a tighter landing cloud, and greater error tolerance, are not four separate effects. They all follow from one property, and that property can be stated deterministically without any Monte Carlo sampling. For a nominal shot, define the landing-position sensitivity as the size of the partial derivative of the landing point with respect to a launch error, $|\partial x_{\text{land}}/\partial v_0|$ and $|\partial x_{\text{land}}/\partial \theta_0|$, computed by finite differences on the deterministic trajectory. Figure 11 plots both against spin. Both fall monotonically. A $1\,\text{m}\,\text{s}^{-1}$ launch-speed error shifts the landing by 0.354 m at 0 RPM but only 0.136 m at 6000 RPM, a 62% contraction, and the launch-angle sensitivity falls by 38% over the same range.

The rest of the paper's stabilisation results follow from this. Because the map from launch error to landing error contracts as spin rises, any distribution of launch errors, whether from turbulence (Sections 4.4 to 4.6) or from human execution (Section 4.10), is squeezed more tightly at the landing end. The reduced variance, the lower entropy, and the improved success rate are all downstream of this one deterministic fact. In the local-sensitivity sense used here, topspin makes the launch-to-landing map progressively less sensitive, so a given launch error produces a smaller landing error as spin rises. That is the clearest single statement of why spin stabilises.

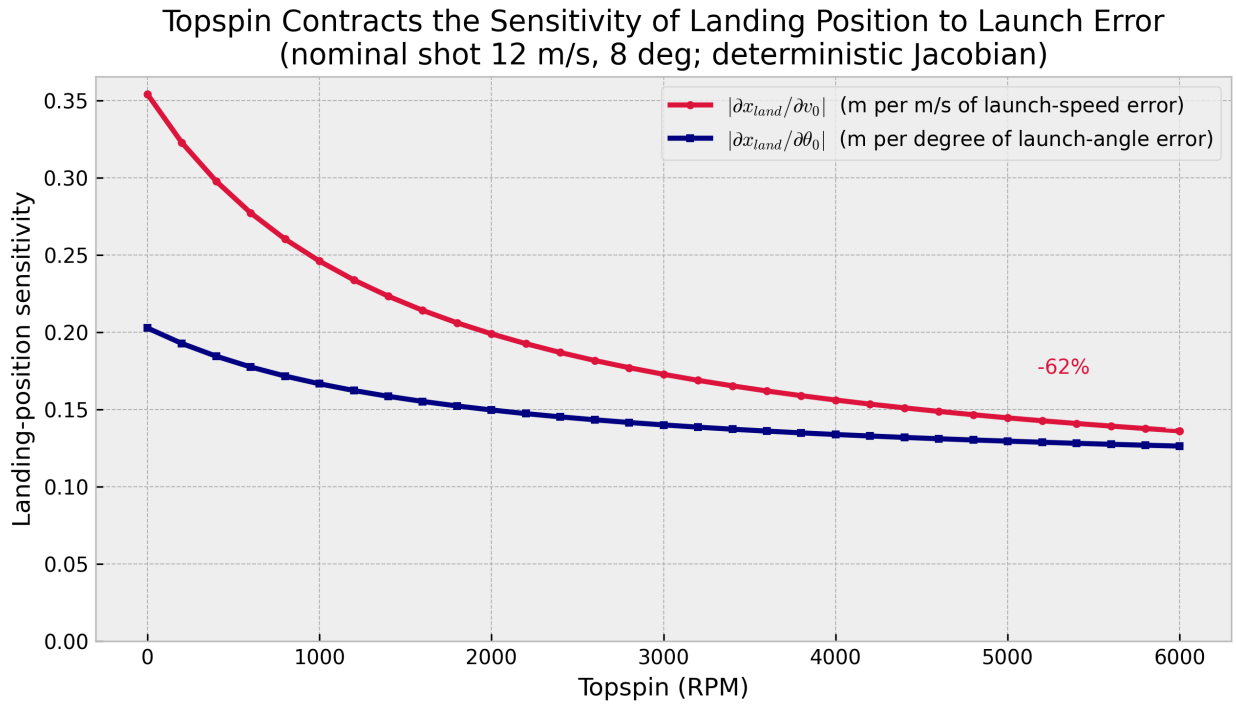



**Figure 11:** Landing-position sensitivity to launch error against spin for a nominal $12\,\text{m}\,\text{s}^{-1}$, 8° shot, computed as a deterministic finite-difference Jacobian. A $1\,\text{m}\,\text{s}^{-1}$ launch-speed error shifts the landing by 0.354 m at 0 RPM but only 0.136 m at 6000 RPM (a 62% contraction); the launch-angle sensitivity falls by 38%. Both curves decrease monotonically.

## 4.12 Statistical Convergence

Because the stochastic results rest on finite Monte Carlo samples, I check that the reported estimates have converged and quote their uncertainty. Figure 12 shows the safe-landing rate at $\lambda = 0.5$ as the sample size grows from 25 to 1600, with 95% Wilson confidence intervals. The intervals narrow as $N^{-1/2}$ and the

estimates settle to about 91% for the loop and about 44% for the flat drive (the 500-shot run of Section 4.10 gives 45%, consistent within Monte Carlo sampling). All figures use a fixed random seed, so every reported value is reproducible under the same implementation.

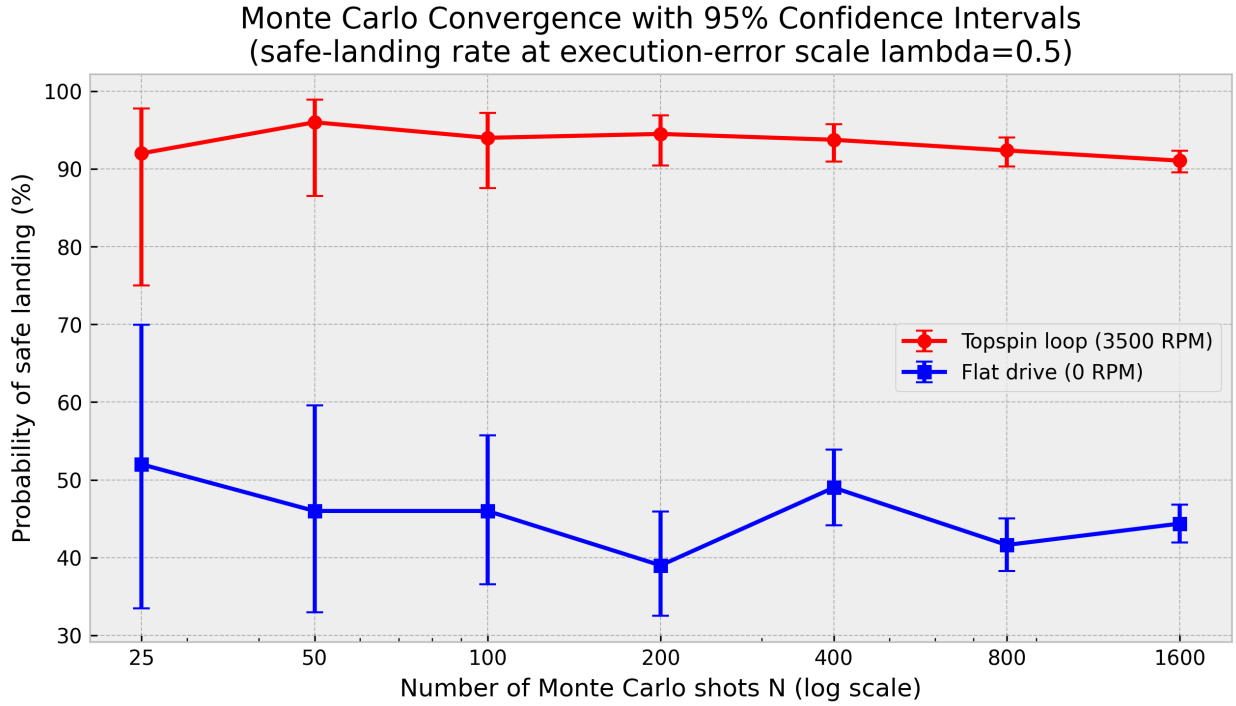



**Figure 12:** Monte Carlo convergence of the safe-landing rate at $\lambda = 0.5$ as the sample size $N$ grows, with 95% Wilson confidence intervals. The intervals narrow as $N^{-1/2}$ and the estimates settle to about 91% (loop) and about 44% (flat drive) by $N = 1\,600$.

## 4.13 Sidespin and Curved Serves: A Three-Dimensional Extension

The model so far lives in the vertical plane, which captures topspin and backspin but not sidespin. Moving to three dimensions generalises the same lift model to an arbitrary spin-axis orientation. The spin becomes a vector $\boldsymbol{\omega} = (0, \omega_{\text{side}}, -\omega_{\text{top}})$, and the Magnus force keeps its general form $\mathbf{F}_{\text{magnus}} = \frac{1}{2}\rho A C_L(S)\, v^2\, (\boldsymbol{\omega} \times \mathbf{v})/|\boldsymbol{\omega} \times \mathbf{v}|$, now with a component out of the vertical plane. With $\omega_{\text{side}} = 0$ the three-dimensional integrator reproduces the two-dimensional trajectories to machine precision, a self-test I enforce, so nothing already reported changes. Sidespin simply adds a lateral force. Figure 13 shows the top-down view of a serve struck with pure sidespin. A ball that flies straight with no spin curves progressively as sidespin is added, deflecting sideways by about 20 cm at 2 000 RPM and 27 cm at 6 000 RPM by the time it reaches the far side. The deflection saturates: trebling the spin from 2 000 to 6 000 RPM adds only about 35% more curve, for the same reason the descent-angle benefit saturated in Section 4.3, because in this model the adopted lift coefficient saturates. This reproduces, from first principles, the hooking serve and sidespin loop every player recognises, and shows the stabilisation framework extends naturally to the full three-dimensional shot.

## 4.14 Optimal Spin for Maximum Error Tolerance

If more spin always helped, the advice would simply be to spin as hard as possible. The model shows that this is wrong at moderate pace. For a fixed launch speed I swept the topspin and, for each value, measured

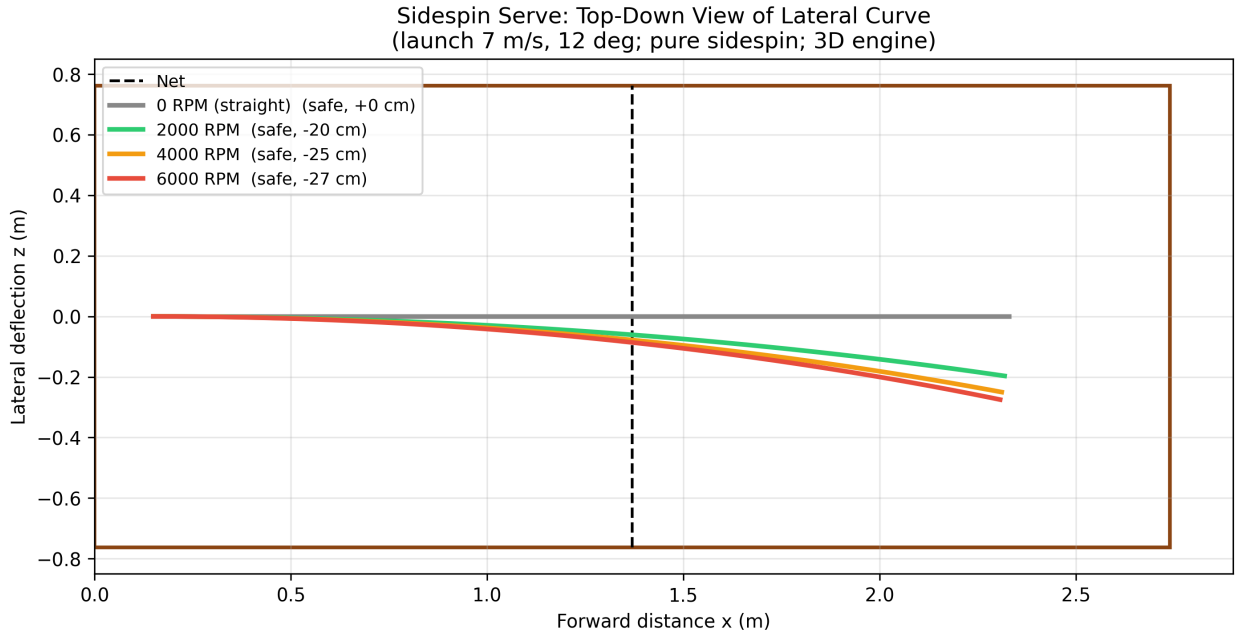



**Figure 13:** Top-down view of a serve ($7\,\mathrm{m\,s^{-1}}$, $12^\circ$) struck with pure sidespin, from the three-dimensional engine. No spin flies straight; sidespin curves the ball sideways, deflecting it about 20 to 27 cm across the table. The curve saturates with spin because the lift coefficient does. The engine reduces exactly to the two-dimensional model when sidespin is zero.

the safe-landing probability under realistic execution noise ($\lambda = 0.4$). Figure 14 (left) gives the result. At $10\,\mathrm{m\,s^{-1}}$ the safe-landing rate peaks near 1 600 RPM and then falls, because over-spinning a slow ball makes it drop so short that a small under-hit catches the net. The optimum is interior, not at the maximum. As pace increases the peak shifts rightward and eventually pins to the ceiling: the error-minimising spin rises from about 1 600 RPM at $10\,\mathrm{m\,s^{-1}}$ to saturation near 8 000 RPM by $13.5\,\mathrm{m\,s^{-1}}$ (Figure 14, right); for this optimisation the spin search was extended to 8 000 RPM, beyond the 6 000 RPM span of the phase-space sweep. This puts a number on the coaching instinct that spin and speed have to be matched. A controlled loop wants about as much spin as its pace calls for, and no more. It also turns a qualitative claim into a prescription: within the simulated speed and spin range, each launch speed has a spin rate that maximises the simulated safe-landing probability.

## 4.15 Validation Against Real Recorded Shots

Every result up to this point is simulated, so the real test is whether the model can predict actual shots it has never seen. For this I used the public SpinDOE dataset (Gossard et al., 2023), which records table tennis ball trajectories together with an accurately measured spin obtained from a printed-dot method. From it I kept the 133 trajectories that carried a measured spin value and a long enough tracked arc, at least six airborne frames before the first bounce. The measured spin is what makes the strong test possible, because it lets the model run in prediction mode rather than fitting mode.

The test was kept deliberately strict. For each of 133 recorded shots I read the launch speed and angle from only the first few frames of the flight, took the spin straight from the SpinDOE measurement, and then let the model predict the whole rest of the trajectory with nothing at all fitted to the later points. That blind prediction was then compared against the recorded flight.

For each shot the prediction error is the root-mean-square distance between the predicted and recorded

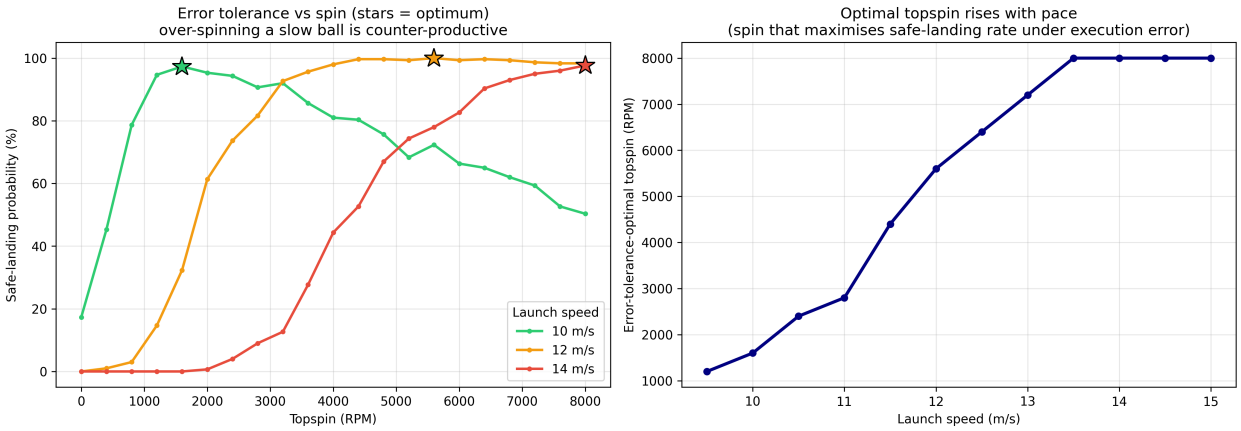



**Figure 14:** Left: safe-landing probability against spin under execution error ($\lambda = 0.4$) for three launch speeds; stars mark the optima. At $10\,\mathrm{m\,s^{-1}}$ the optimum is interior, and beyond it more spin lowers the success rate. Right: the error-minimising topspin rises with launch speed, from about 1 600 RPM at $10\,\mathrm{m\,s^{-1}}$ to the roughly 8 000 RPM search ceiling by $13.5\,\mathrm{m\,s^{-1}}$.

ball positions over all tracked frames, and $R^2$ is the coefficient of determination between predicted and recorded positions; no model parameters were tuned to these trajectories. Across these 133 shots, spanning topspin and backspin from about $-2\,900$ to $+4\,700$ RPM, the median prediction error was 37.8 mm over the entire arc, with a median $R^2$ of 0.994. In plainer terms, on a typical shot the predicted path stayed within about four centimetres of the real ball across a flight of two to three metres, using the measured launch state and spin with all physical parameters fixed in advance. Figure 15 shows a representative topspin and a representative backspin shot, both at that median error, with the predicted curve laid over the recorded points. Just over half the shots came in under five centimetres and about two thirds under eight. The larger errors clustered in shots with strong sidespin or very short tracked arcs, which a model living in a single vertical plane cannot fully capture, and which points straight at the three-dimensional extension of Section 4.13 as the next step.

Every stabilisation figure in the paper comes from this same engine, and here that engine predicts real, independently measured flights to a median error of under four centimetres without being tuned to them. The model is not only internally consistent; it reproduces the measured trajectories.

# 5 Discussion and Coaching Implications

The results settle into one picture: topspin makes shots faster, more consistent, and more forgiving of execution error at the same time. Table 3 maps these findings onto specific named shots.

For beginners, the phase-diagram result is the most actionable. At 0 RPM the safe window spans only about $2\,\mathrm{m\,s^{-1}}$ of swing speed, and by 3 000 RPM it roughly doubles. The single most important technical goal at this level is to generate topspin by brushing the ball up and forward rather than hitting it flat. Even an imperfect, poorly timed forehand loop widens the margin for error before technique has actually improved.

For club-level players, the message for windy or draughty halls is simple: loop the ball, but do not feel you have to spin it as hard as you can. Looping clearly beats hitting flat, because a loop still lands on the table where a flat drive of the same pace would sail long. Among loops, the extra stability from more spin flattens out: a committed loop of two or three thousand RPM already gives most of the benefit, and spinning harder than that adds little. So commit to a controlled loop for the margin it gives, and put the extra effort into placement rather than into squeezing out maximum spin.

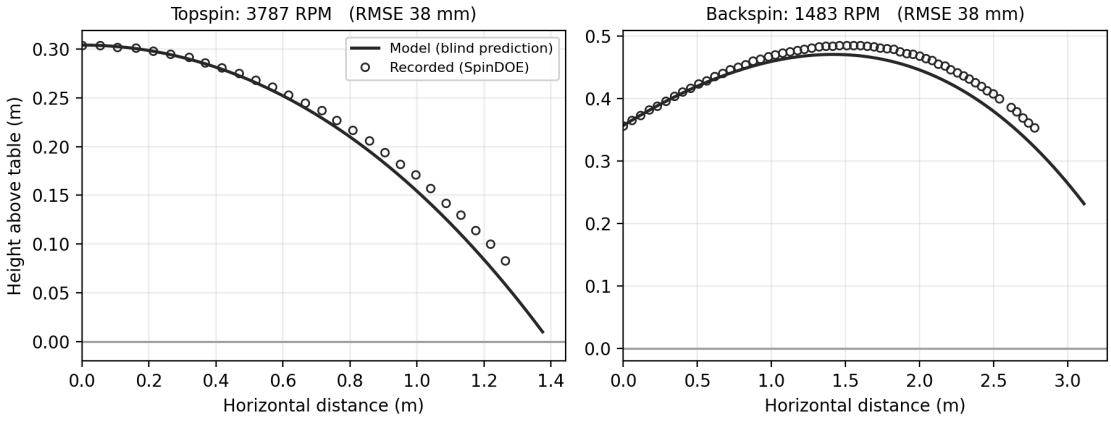



**Figure 15:** Blind prediction against real recorded shots from the SpinDOE dataset (Gossard et al., 2023). For each shot the launch speed and angle are read from the first few frames, the spin is the dataset's measured value, and the model then predicts the entire remaining flight with nothing fitted to it (solid line), shown against the recorded ball positions (open circles). Both examples sit at the dataset's median per-shot error (RMSE 38 mm); across all 133 shots the median $R^2$ is 0.994.

For intermediate players, the contact-height result is directly usable. Taking the ball lower and earlier, in the rising phase just after the bounce, raises the safe-speed ceiling and amplifies the Magnus benefit per RPM. Players should develop a banana flick or backhand flick return of serve and learn the forehand loop on the rise.

For advanced players, the entropy result matters most. A player sustaining 5 000 to 6 000 RPM on their loops can direct the ball to the opponent's elbow or crossover point with a consistency a flat drive cannot match. Placement precision is the one payoff that keeps growing past a few thousand RPM: simply keeping the ball on the table saturates early (Section 4.4), but the landing distribution keeps tightening as spin rises (Section 4.6). The loop kill and sidespin loop serve are the shots where this placement edge is most exploitable. In the third-ball attack, maximising spin against a heavy backspin return is well justified, because the shallower descent from the chop has to be overcome with extra Magnus force.

**Table 3:** Named table tennis shots, representative spin ranges, and the primary aerodynamic benefit quantified in this study.

| Shot | Spin (RPM) | Primary benefit |
|---|---|---|
| Flat drive | 0–500 | Speed only; narrow safe window |
| Topspin loop | 2000–4000 | Safety margin: +41 to 61% ceiling |
| Loop drive | 3000–5000 | Speed combined with safety |
| Loop kill | 4000–6000 | Placement precision (entropy −34%) |
| Banana / BH flick | 2000–3500 | Early-ball timing, low contact |
| Chop (backspin) | 0–2000 | Flatter descent; hard to attack |
| Push (backspin) | 0–1000 | Low trajectory; forces hard return |

# 6 Conclusions

This study showed that topspin works as a multi-layered aerodynamic stabiliser. It is not merely a way to curve the ball. It widens the margin for pace, sharpens the descent, and quietly reduces how much the errors a player is bound to make move the ball.

Geometrically, topspin steepens the descent angle by 3.2° and raises the maximum safe launch speed by 61% at 6000 RPM. These are not cosmetic corrections. They mark the line between a shot that is mechanically possible and one that is not. At $14\,\mathrm{m\,s^{-1}}$ with no spin, none of the launch angles tested produced a safe landing, since the no-spin ceiling is only $9.2\,\mathrm{m\,s^{-1}}$, whereas at 6000 RPM the 8° case clears a much wider band of speeds.

Statistically, topspin cuts the mean squared landing displacement by 81% and the Shannon entropy of the landing distribution by about 34%. A player using heavy topspin produces a fundamentally more predictable flight, with roughly half as many effectively distinct landing outcomes.

These predictions are not only internally consistent. Tested against 133 real recorded shots from a public dataset, the same engine reproduces measured flights to a median of under four centimetres over the whole arc, from the measured launch and spin. The physics behind the coaching advice holds up against the real ball, not just against itself.

The most direct coaching implication comes from the human-error result. At moderate execution error a topspin loop lands safely 91% of the time against 45% for a flat drive. Because both shots receive the same error distribution, the gap between these two configurations comes from their flight dynamics rather than from any difference in consistency: two players of identical consistency, one looping and one hitting flat, would produce very different error rates under match pressure.

Table tennis looks like a game of reactions, fast hands, quick feet, sharp eyes. At the level of the ball, it is a problem of aerodynamic control under uncertainty: how do you get a 2.7 g object moving at 50 km/h to land inside a 2.74 m window when air turbulence and human inconsistency are both working against you? The answer, in this study and in the game itself, is spin. Every added RPM of topspin hands a little more of the job from the player's nervous system to the laws of fluid dynamics. Learning to loop is not only a technical improvement, it is a smarter way to play the game.

## 6.1 Limitations

The core analysis is two-dimensional, although the three-dimensional extension of Section 4.13 reproduces sidespin and lateral curve. Air density is held constant through the flight, and bounce dynamics are excluded, so the model covers only the airborne phase. The engine is verified against analytic limits, its own physical self-tests, and 133 real recorded trajectories (Section 4.15). What it has not yet been tested against is ordinary broadcast or phone footage of live match play, where the ball is not marked and spin has to be inferred rather than measured directly. The larger prediction errors on the real dataset also came mostly from strong sidespin, which the two-dimensional model cannot fully represent. The lift law is a smoothly saturating one, so it does not reproduce the non-monotonic "lift crisis" that flight experiments have found for table tennis balls at intermediate spin (Miyazaki et al., 2017), nor the inverse Magnus force reported at very low spin; a more detailed lift curve fitted to that data is a natural refinement. The spin-decay rate is taken as a single representative value, within the range measured for table-tennis flight (Ziegler et al., 2023) but not fitted to the specific ball and surface used here.

## 6.2 Future Work

With validation against a marked-ball dataset in hand, the next step is to repeat the comparison on unmarked footage from real matches, where spin must be recovered from the trajectory itself rather than read from a printed dot. Further extensions include combined spin (simultaneous top- and sidespin, as in the hook loop), which the real-data errors suggest matters most; the aerodynamics of the bounce; and turning the optimal-spin result of Section 4.14 into a control law across a wider range of match conditions.

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